

# Muhammad Abdullah

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## EDUCATION

**Ghulam Ishaq Khan Institute of Engineering Sciences and Technology**  
*B.Sc. in Cyber Security; GPA: 3.42/4.00*

Swabi, Pakistan  
Sep 2022 – current

## SKILLS

**Penetration Testing:** Metasploit, Bettercap, Burp Suite, Nmap, Aircrack-ng, Hydra, John the Ripper, SQLMap  
**Network Security:** Wireshark, Snort, Suricata, Zeek, ARP Spoofing, DNS Poisoning  
**Programming:** Python (Offensive Security Scripting), Bash, C++  
**Red Teaming:** Social Engineering, Privilege Escalation, Post-Exploitation Techniques  
**Reverse Engineering:** Ghidra, IDA Pro, Radare2  
**Digital Forensics:** Volatility, Autopsy, FTK Imager  
**Operating Systems:** Kali Linux, Parrot OS, Windows Security Hardening  
**Other:** Git, Docker, Flask, SQL, Virtualization (VMware, VirtualBox)

## EXPERIENCE

**Vulnerability Assessment and Penetration Testing (VAPT) Intern**  
*National Cyber Emergency Response Team (NCERT / PKCERT)*

July 2025 — August 2025

- Completed a 6-week internship focused on exploitation and security assessment of Windows 7, 10, and 11 environments.
- Researched and demonstrated antivirus and EDR bypass techniques through payload encryption, obfuscation, and loader development.
- Conducted web application penetration testing on internal PKCERT platforms, identifying and documenting high-impact vulnerabilities.
- Utilized tools such as Burp Suite, Metasploit, Wireshark, and custom scripts for exploitation and analysis.

**Capture The Flag (CTF) Player**

2023 — Present

*National and International Cybersecurity Competitor*

- Competed in CTFs including Ignite National Cybersecurity Hackathon, Pakistan Cybersecurity Challenge (PCC) by NCCS, Hack The Box, and BlackHat MEA Online Qualifier (185th out of 1500 teams).
- Focused on offensive security domains such as penetration testing, reverse engineering, binary exploitation, and web security.
- Collaborated with teams to solve advanced challenges, enhancing red teaming, exploit development, and vulnerability research expertise.

## AWARDS & ACHIEVEMENTS

**Dean's Honor List:** Awarded by the Dean Faculty of Computer Science and Engineering for obtaining GPA above 3.50 in 1st, 3rd, and 4th semesters. (Fall 2022, Fall 2023, Spring 2024)

**Top 10/140 in Ignite National Cybersecurity Hackathon – Regional Round:** Secured a top 10 position in the Islamabad/Rawalpindi regional round of the Ignite National Cybersecurity Hackathon 2024, competing against elite cybersecurity teams. (Dec 2024)

**Nominated for Intellectual Student Award – GIK Institute:** Recognized for outstanding academic performance and consistent high grades throughout the academic career. (2025)

**2nd Position – Federal Board STEAM Projects Competition (Physics):** Achieved 2nd place in the Intermediate STEAM Projects competition, Federal Board, for developing a remote-controlled car using Arduino Uno and embedded programming. (2022)

### **Stealthy Hardware-Assisted Malware Development (Final Year Project)**

- Research and development of a hardware-assisted offensive security prototype to evaluate enterprise detection and response capabilities for red-team operations and offensive research.
- Architecture of the solution to establish resilient, persistent access for reconnaissance, lateral movement and controlled post-exploitation testing.
- Implementation of advanced evasion techniques like payload encryption, multi-staged payloads, runtime decryption, in-memory execution, process hollowing, process doppelganging, process injection, AMSI/ETW bypass, code obfuscation, junk code, packing etc.
- Integration of Command and Control servers(C2) and disguises such as USB chargers or USB storage devices like Rubber Ducky for covert deployments.

### **Operation Redbridge – Digital Forensics Case Study | [GitHub](#)**

- Conducted an in-depth digital forensics investigation on a simulated sexual assault case involving concealed and encrypted digital evidence.
- Analyzed exhibits including a steganographic image, high-entropy files, and damaged storage devices using Autopsy and open-source tools.
- Recovered encrypted messages (Base64 + Vernam cipher), drug-related search history, and aggressive chat logs supporting the timeline of the incident.
- Documented evidence indicating digital obfuscation, suspect behavioral profiling, and potential criminal planning.

### **Intrusion Detection System using Machine Learning | [GitHub](#)**

- Developed a machine learning-based Intrusion Detection System (IDS) trained on the Botnet Dataset 2010.
- Implemented classification models to detect network anomalies and identify potential cyber threats with high accuracy.
- Optimized feature selection and model performance for real-time network traffic analysis.

### **ARP Spoofer | [GitHub](#)**

- Created an ARP spoofing tool in Python to conduct MITM (Man-in-the-Middle) attacks for penetration testing purposes.
- Implemented network packet interception and redirection techniques to manipulate traffic.
- Developed countermeasures to detect and prevent ARP spoofing attacks.

### **Stealth Keylogger | [GitHub](#)**

- Designed a Python-based stealth keylogger for monitoring keystrokes in controlled cybersecurity testing environments.
- Implemented encrypted log storage and stealth mechanisms to evade detection.
- Designed evasion techniques to bypass common security measures and enhance persistence.